Hongyuan Xu

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Education

Northwestern University Dec 2026 (Expected)
Master of Science in Computer Science • GPA 3.75

University of Massachusetts Amherst Sep 2020 – May 2024
Bachelor of Science in Computer Science • GPA 3.74 • Dean's List

Technical Competencies

Robotics: Robot Manipulation, Robot Locomotion, RL (DDPG/PPO/SAC), PID, SLAM, Sensor Fusion
Simulation: Isaac Lab, MuJoCo, Gazebo, Carla, Domain Randomization, Sim-to-Sim, Sim-to-Real
Compute: Parallel RL Training, Multi-GPU Workflows (8× NVIDIA B300 GPUs)
Software: Python, C++, ROS/ROS2, PyTorch, Linux, Docker, Git, CI/CD

Experience

Xenobot / Living Machines Lab — Research Assistant (RL & Modular Robotics) Nov 2025 – Present

- Developing reinforcement-learning controllers in Isaac Lab for reconfigurable modular legs, training locomotion primitives across multiple morphologies.
- Building sim-to-real workflows for physical modular legs through actuation calibration, sensor-noise modeling, and domain randomization.

Northwestern University Robotics Lab — Graduate Research Assistant Jan 2025 – Nov 2025

- Architecting a multi-robot monitoring backend streaming voltage, command queues, and telemetry logs for real-time fleet diagnostics.
- Built a Raspberry Pi–based base-station "air-traffic-controller" fusing OptiTrack streams and broadcasting high-level mode commands (take-off, hover, land, e-stop) to 10+ drones.

LightWheel AI — End-to-End Self-Driving Algorithm Developer Aug 2024 – Nov 2024

- Built simulation pipelines integrating nuScenes and nuPlan datasets to evaluate SOTA autonomous driving models in both open-loop and closed-loop settings.
- Automated benchmarking for 2,000+ scenarios by feeding camera frames into end-to-end models and comparing outputs to ground truth labels.
- Contributed to scaling UniAD and VAD training from nuScenes to nuPlan, improving model generalization and reliability.

Projects

ARM_REBOT: Modular 6-DOF Robot Arm — Onshape, Isaac Lab, PyTorch Jan 2026 – Present

- Designed the modular 6-DOF arm in Onshape; built it with swappable motors, 3D-printed brackets, carbon-fiber rods, and a rack-and-pinion gripper.
- Rebuilt 535 CAD parts into a 10-body Isaac Sim model; validated joint motion and gripper force, informing a motor upgrade.
- Trained PPO in 1,024 parallel GPU environments: 2.2 mm median error; 97.7% of 1,536 targets within 1 cm.
- 3D-printed components and tested motors on hardware; preparing sim-to-real transfer of the trained policy.

Universal Robot Locomotion Controller — MuJoCo Jan 2026 – Present

- Developed a model-based teacher–student controller that uses sensorimotor history to replan among 4,096 CPG motion primitives on unseen topologies.
- Parallelized training on an 8-GPU NVIDIA B300 server; evaluated 3,000 unseen morphologies in MuJoCo, outperforming tested controllers across all three module sizes.

Autonomous Robotic System for Target Object Search — Python, ROS, RVIZ Jan 2024 – Jun 2024

- Built a ROS software stack integrating Triton robot sensors (camera, LiDAR, IMU) for autonomous mapping and exploration.
- Designed SLAM-based exploration policies and generated real-time local maps during navigation.
- Transformed global coordinates to robot-local LED indicators for intuitive target localization feedback.